Zachary Bakhchi

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Education

The Cooper Union for The Advancement of Science and Art — New York, NY
B.E. Electrical Engineering, Half-Tuition Scholarship — *Aug. 2023 – May 2027*
Coursework: Data Structures & Algorithms, Hardware Design, Computer Networks, Operating Systems, Software Engineering, Signal Processing, Digital Signal Processing, **Medical Imaging**, **Frequentist Machine Learning**, **Cybersecurity**, **ML for Medical Imaging**

Experience

The Cooper Union for The Advancement of Science and Art — New York, NY
Teaching Assistant – CS-291, Agentic Coding — *Sept. 2026 – Present*
- Tested and refined agentic coding lessons and exercises before deployment to students.
- Mentored students in need of direction or assistance.

The Conference Board — New York, NY
AI Intern — *June 2026 – Aug. 2026*
- Developed LLM agents and automated workflows for marketing, training, and business applications.
- Built AI-powered dashboards, forecasting tools, and data-driven applications for internal teams.
- Re-engineered a legacy chatbot using modern LLM architectures and expanded its capabilities.

Effy Jewelry — New York, NY
IT Intern — *June 2026 – Aug. 2026*
- Developed AI and automation tools using PowerShell, vendor APIs, and Model Context Protocol (MCP).
- Built internal CRUD applications, including input from corporate leadership and technical stakeholders.
- Modernized enterprise IT workflows, payment processing security, incident response, and user onboarding processes.

Ana's Cloud LLC — Manhasset, NY
Software & Systems Engineer — *Sept. 2023 – May 2026*
- Promoted from Intern to Software & Systems Engineer, supporting enterprise IT, cybersecurity, and AI initiatives.
- Developed a managed intrusion detection platform and AI chatbot for a major client.
- Resolved hundreds of support incidents, maintained client environments, and executed cloud migrations.
- Evaluated Microsoft Copilot and LLM solutions while maintaining network and infrastructure documentation.

Selected Projects

Spectral Domain Optical Coherence Tomography (SD-OCT) | *MATLAB, Signal Processing* — *April 2026*
- Developed a vectorized SD-OCT pipeline to reconstruct 1D A-Scans and 2D B-Scans from raw interference data.
- Implemented background subtraction, spectral windowing, polynomial deconvolution, and artifact suppression.
- Built a Spectral Domain Phase Microscopy (SDPM) algorithm to track sub-nanometer membrane vibrations.

CooperLink | *Java, React, Azure* — *May 2026*
- Built a full-stack platform connecting Cooper Union students with alumni mentors and career opportunities.
- Developed Spring Boot/PostgreSQL backend and React frontend with Firebase authentication.
- Containerized with Docker Compose and deployed on Azure.

MRI Compressed Sensing Reconstruction | *MATLAB, Image Processing* — *May 2026*
- Developed an MRI reconstruction pipeline using compressed sensing and wavelet-domain sparsity.
- Implemented Fourier analysis, multilevel wavelet decompositions, and random undersampling.
- Built an iterative reconstruction algorithm (ISTA) to evaluate image recovery and reconstruction error.

Skills

Technical Areas: Full-Stack Development, AI & Machine Learning, Cybersecurity, Signal & Image Processing
Languages: Python, Java, JavaScript/TypeScript, SQL, C/C++, MATLAB
Frameworks & Tools: React, Node.js, Spring Boot, OpenCV, Docker, Git/GitHub, Azure, Firebase, PostgreSQL
Platforms: Linux, Windows

Leadership & Activities

Leadership: President of Cooper Union Hillel **Organizations**: Cooper Union IEEE